

Loh Boon Cheong, Timothy

Singapore • lohbtimothy@gmail.com • +65 9726 6176

Portfolio • GitHub • LinkedIn

Seeking a one-year software engineering internship: May 2027 – May 2028

Education

Singapore Institute of Technology | DigiPen Institute of Technology Singapore

BSc in Computer Science in Real-Time Interactive Simulation

Expected Graduation: 2028

GPA: 4.0

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Networking, Graphics Programming

Singapore Polytechnic

Diploma in Accountancy; GPA: 3.0

Graduated: 2022

Projects

Hellmaker 2D Engine/Editor

September 2025 – May 2026

Technical Lead / C++, Lua, OpenGL, Dear ImGui, FMOD

- Designed and implemented the engine's system registrar and lifecycle management, integrated subsystems into the shared engine/editor framework, and built an embedded web server for browser-based runtime diagnostics.
- Resolved video/audio synchronization drift by using FMOD's audio clock to drive video playback, accounting for output latency and coordinating decoded audio through a mutex-protected sample queue.
- Implemented a PNG-to-BC3/DXT5 import pipeline and direct compressed texture uploads to OpenGL, improving editor and game loading times.

StrokeNet – Networked Drawing Game

March 2026

Architecture Lead / C++, Winsock, SFML

- Designed data flow between threads in both the client and server, synchronization boundaries around shared state, and binary packet layouts for network commands in a four-person LAN drawing-and-guessing game.
- Designed a UDP protocol combining acknowledged requests and notifications with best-effort drawing updates, using non-blocking Winsock sockets.

Interactive Graphics Demos

January 2026 – May 2026

C++, GLSL, OpenGL, GLFW, GLEW, GLM

- Implemented nine interactive coursework demos within a supplied graphics framework, covering ray casting, ray tracing, Blinn-Phong lighting, environment mapping, portals, and deferred rendering.
- Built a six-pass cartoon rendering pipeline combining shaded scene output, Gaussian blur, Sobel edge detection, and compositing; implemented normal mapping and G-buffer surface data for deferred shading.

x86-64 Assembly Mini Projects

August 2026 - Current

GCC, x86-64 Assembly, Shell, Linux

- Implemented multiple square-root approximations and integrated the C standard library's sqrt through an assembly wrapper, applying System V x86-64 calling conventions, register preservation, and stack alignment.
- Created a shell-based benchmark harness using GCC and perf to compare executable runtimes across shared test inputs.

Area-Preserving Polygon Simplification

May 2026 – June 2026

Implementation / C++, Python, Make

- Implemented greedy segment collapse using circular linked rings, a min-heap, and a spatial grid to reduce polygon vertex counts while checking intersections and preserving signed area within floating-point precision.
- Used generation counters to reject stale candidates and lazy edge pruning for spatial queries; evaluated behavior with supplied and additional test cases. Developed with documented AI assistance.

Technical Skills

Languages: C, C++, C#, Python, Lua, SQL, GLSL

Tools: GCC, x86-64 Assembly, Git, CMake, Make, Visual Studio, Jira, Batch Scripting

Libraries and APIs: OpenGL, GLFW, GLEW, GLM, Dear ImGui, SFML, Winsock, FMOD

Core Areas: Data Structures and Algorithms, Multithreading, Memory Management, Networking, Object-Oriented Programming, Computer Architecture, Shader Programming

Additional Experience

Foo Kon Tan LLP

Accounting Intern

September 2021 – November 2021

- Supported daily accounting operations through data entry, account reconciliations, and financial report preparation.
- Assisted with invoice processing and documentation, maintaining accurate records and supporting compliance with accounting standards.